

SEC2-FA4_GROUP25_DICHE,J.

<https://github.com/DyasMine-MD/APM1110---FA/blob/main/FA4>

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#2.Binary channel calculations
P_tx0 <- 0.70
P_tx1 <- 0.30
P_0_to_1 <- 0.05 # transmitted 0 but received 1
P_1_to_1 <- 0.75 # transmitted 1 and received 1

# (a) Probability a 1 was received
P_recv1 <- P_tx0 * P_0_to_1 + P_tx1 * P_1_to_1

# (b) Probability a 1 was transmitted given a 1 was received (Bayes' rule)
P_tx1_given_recv1 <- (P_tx1 * P_1_to_1) / P_recv1

# Print results
cat("Binary channel results\n")

## Binary channel results
cat(sprintf("P(received 1) = %.3f (%.1f%%)\n", P_recv1, 100 * P_recv1))

## P(received 1) = 0.260 (26.0%)
cat(sprintf("P(transmitted 1 | received 1) = %.4f (%.2f%%)\n",
            P_tx1_given_recv1, 100 * P_tx1_given_recv1))

## P(transmitted 1 | received 1) = 0.8654 (86.54%)

# Employees and error rates
share <- c(Jane = 0.10, Amy = 0.30, Ava = 0.60)
err_rate <- c(Jane = 0.08, Amy = 0.05, Ava = 0.01)

# Overall error probability (law of total probability)
P_error <- sum(share * err_rate)

# Posterior probabilities P(author | error) via Bayes' rule
posterior <- (share * err_rate) / P_error

# Print results
cat("Employees and error results\n")

## Employees and error results
cat(sprintf("Overall error rate = %.4f (%.2f%%)\n\n", P_error, 100 * P_error))

## Overall error rate = 0.0290 (2.90%)
```

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cat("Posterior probabilities given an error:\n")

## Posterior probabilities given an error:
print(round(posterior, 4))

##   Jane    Amy    Ava
## 0.2759 0.5172 0.2069
cat(sprintf("\nMost likely author given an error: %s (%.2f%%)\n",
           names(which.max(posterior)), 100 * max(posterior)))

##
## Most likely author given an error: Amy (51.72%)

```